

Lightweight Tuberculosis Screening Pipeline for Community-based Active Case Finding

Irish Danielle Morales , Raphael B. Alampay , Ruben A. Saulog  and Patricia Angela R. Abu 
Department of Information Systems and Computer Science, Ateneo de Manila University, Quezon City, Philippines

Abstract—Tuberculosis (TB) screening is widely done with computer-aided detection (CAD) systems. While TB CAD systems are highly accurate, community-based active case-finding (ACF) interventions face barriers in connectivity, compute, and interpretation. ACF often conducts high-volume screening in remote community locations where compute and connectivity are limited. The TB CAD system then reads each chest X-ray on-site for presumptive TB and outputs probability scores and visual overlays of abnormalities. However, these CAD outputs are primarily designed for radiologist reading, whereas ACF field teams often do not involve a radiologist and on-site readers without radiology training evaluate the CAD results. We present a TB screening pipeline that pairs lightweight classification and object detection models with a natural-language description of model findings. For classification we introduce FlipR, a lightweight symmetry-gated network that achieves comparable performance to the strongest baselines despite using the fewest parameters. The classifier achieves performance above the World Health Organization's standards for specificity and sensitivity in TB CAD. For object detection we introduce a custom backbone that increases accuracy of active TB lesions over base YOLO26. Findings are rendered as a deterministic, faithful-by-construction natural-language report following radiologic report convention, so the text asserts only what the models detected. The pipeline targets adult screening and is deliberately lightweight for deployment in resource-limited settings. However, the design and evaluation of the system was constrained by limited TB-specific samples in public datasets. Further work remains to evaluate deployment feasibility, detect specific radiographic findings, and evaluate natural language outputs through a radiologist reader study.

Keywords—Tuberculosis; Computer-Aided Detection; Chest X-ray; Medical Imaging; Community-based Active Case Finding

I. INTRODUCTION

Tuberculosis (TB) causes more deaths than any other infectious disease worldwide [1]. It is caused by airborne *Mycobacterium tuberculosis* and primarily affects the lungs, presenting as pulmonary TB [3]. Pulmonary TB is typically diagnosed by assessing patient history of symptoms, chest radiography, and bacteriological confirmation. In a clinical setting, the patient will be interviewed for symptom history by a clinician, typically a general practitioner or pulmonologist. The clinician may then request a chest X-ray (CXR) of the patient. A radiologist interprets CXRs for abnormalities indicative of TB and reports their clinical impression in a radiographic report. If patient symptoms or radiographic report indicate TB, the patient is said to have presumptive TB. Bacteriological confirmation is needed to confirm TB, and is done through smear microscopy, culture, or rapid diagnostics such as Xpert MTB/RIF or Truenat. Once confirmed, the patient is said to have bacteriologically confirmed TB. Upon clinical diagnosis of TB, the patient then begins treatment.

Community-based active case finding (ACF) refers to an intervention where teams screen at-risk populations in community sites or through home visits, rather than waiting for symptomatic patients to present at a clinic [2]. Figure 1 shows a representative ACF workflow, where a field team carries out screening and triage on the community site and an off-site laboratory performs confirmation.

A. Computer-aided Detection of Tuberculosis

TB computer-aided detection (CAD) systems are widely used in both clinical settings and ACF. The World Health Organization (WHO) endorsed CAD systems as an alternative to human readers for patients aged 15 years and older. In 2025, WHO approved six commercial products as meeting its TB screening performance standards [7]. In one reader study, CAD assistance improved radiologist performance and enabled non-radiologists to achieve similar performance as radiologists [9]. Several reviews report CAD model accuracy approach WHO targets across different populations [8], [34], [35], [14].

In ACF, WHO positions TB CAD as a screening and triage tool that selects who proceeds to confirmatory testing [6], [2]. Since CAD systems are used in place of a human reader, community-based ACF teams commonly do not include a radiologist [21], [22]. Despite WHO endorsement, however, clinicians across countries currently value CAD as a complement to, not a replacement for, expert reading [33].

B. Challenges in CAD Integration in Community-based ACF

While TB CAD systems are highly accurate, CAD workflow integration, local validation, and human factors are relatively under-studied [10], [8]. ACF, in particular, faces barriers in connectivity, compute, and interpretation:

Connectivity and compute are constrained in community sites. ACF is often conducted in remote community sites, where connectivity may be unstable and laptops are used to run CAD software [21]. Implementation studies in India and Vietnam document how workflow, connectivity, and interpretation conditions shape CAD deployment in the field [31], [32].

CAD overlays are designed for radiologist readers, whereas ACF teams often do not have radiologists. We review the six TB CAD products approved by WHO in 2025: qXR v4.0 [15], CAD4TB v7 [16], Lunit INSIGHT CXR v3.1 [17], DrAid v2.4.4-6 [18], Genki Edge v3.4.2 [19], and InferRead DR Chest v1.0.1.1 [20] [7]. All deliver TB screening results as a probability score with visual localization through a boxed region, outline, or heatmap. Some products additionally

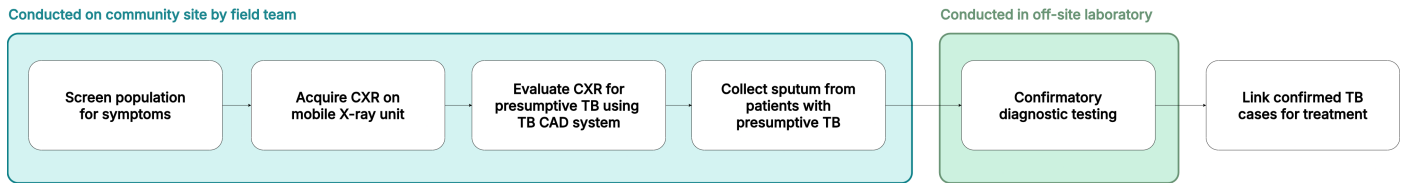


Fig. 1. A TB community active case-finding (ACF) workflow. The CAD system in ACF is often used by a radiographer rather than a radiologist.

generate broader radiology reports spanning many chest findings [18], [15], but such reports are produced for radiologist reading rather than for the non-radiologist readers in ACF, and the TB screening output itself remains a score and an overlay. CAD overlays are also found to overlap poorly with expert-identified regions and do not convey the contextual reasoning used during diagnosis [27]. A TB CAD acceptability study found that some users wanted to understand why the model reached its conclusions [38], and textual descriptions have been proposed as a more clinically meaningful complement to image markup [36], [28], [11], [37]. This presents unique constraints to ACF, which is often carried out by mobile field teams that include a radiographer but often no radiologist on-site [22]. Despite chest X-ray interpretation falling outside the radiographer's scope of practice, the radiographer in ACF decides whether to refer a patient for confirmatory testing using the radiograph, the patient's symptoms, and the CAD output [21].

C. Related Work

TB CAD models with lightweight memory and compute have been explored in previous studies, from LightTBNet at under a million parameters [46] to lightweight hybrid networks [47], building on edge-deployed chest X-ray models for related conditions [25], [26]. Some existing commercial CAD systems require a dedicated CAD box or separate CAD laptop to support offline screening due to compute constraints [21].

Related literature has also explored models that generate natural-language descriptions of findings. In a research prototype, Siemens Healthineers paired a combined classification-localization model [12] with a domain-adapted LLM to generate CXR findings [11]. General medical vision-language models such as LLaVA-Med [13] and CXR-LLaVA [29] demonstrate image-conditioned radiology text generation, and recent surveys catalogue the rapid growth of vision-language models for medical report generation [30]. However, most work remains in general chest X-ray detection, with few specific to tuberculosis.

II. SYSTEM OVERVIEW

This paper presents a lightweight TB screening pipeline that describes model findings in natural language, as shown in Figure 2. A CXR (DICOM or standard image format) is ingested from the screening unit's X-ray or uploaded directly, then scored by the TB classification module. For images flagged as TB, the object detection module identifies active and latent TB regions. The classification and detection outputs are passed to the report generation module, which produces a natural-language report of the findings based on radiologic report convention. All three stages are lightweight, for screening where no radiologist and often no GPU is present.

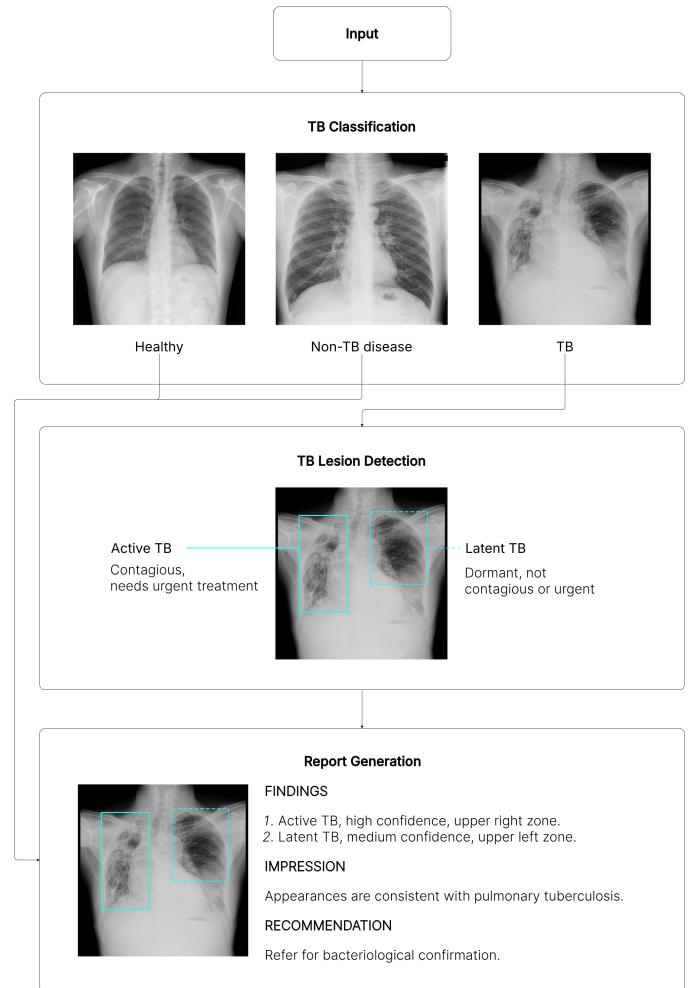


Fig. 2. Overview of the TB CAD pipeline: image classification, TB lesion detection, and report generation.

III. TUBERCULOSIS CLASSIFICATION

The TB classification module flags each CXR as either healthy, sick non-TB, or TB as shown in Figure 2.

A. Dataset

Classification and object detection models are trained on a simplified redistribution¹ of TBX11K [39], a public CXR dataset with image-level labels (*healthy*, *sick non-TB*, *TB*) and bounding-box-level labels (*active-TB*, *latent-TB*).

TBX11K is selected because it is the largest public TB CXR dataset with a TB-specific sample large enough to train

¹<https://www.kaggle.com/datasets/vbookshelf/tbx11k-simplified>

from scratch, expert bounding-box annotations, a sick non-TB class, and a demographic and class composition that reflect real-world clinical settings [40]. TBX11K notes per-image patient age and sex, with most TB cases aged 20 to 60 and a male majority (72% of TB X-rays), which is representative of the general population and consistent with real-world clinical scenarios [40].

The official TBX11K test set is withheld for an online challenge, so we use only the public trainval images. Our results are therefore not directly comparable to SymFormer [40] or other challenge entries on the private test set.

Classification uses the image-level labels shown in Table I. The 8,400 public CXRs (3,800 healthy, 3,800 sick non-TB, 800 TB) are partitioned per class into train, validation, and test folds in a 70%/15%/15% ratio. All reported classification metrics are computed on the held-out test fold of 1,260 CXRs (570 healthy, 570 sick non-TB, 120 TB).

TABLE I. TBX11K DATASET COMPOSITION (IMAGE-LEVEL LABELS) FOR CLASSIFICATION

Class	Train	Val	Test	Total
Healthy	2,660	570	570	3,800
Sick non-TB	2,660	570	570	3,800
TB	560	120	120	800
Total	5,880	1,260	1,260	8,400

Per-class 70/15/15 train/validation/test split.

B. Data Preprocessing

Classification models are trained without preprocessing or data augmentation. TBX11K CXRs are acquired under a standardized posteroanterior protocol and are largely uniform in pose, so aggressive augmentation risks introducing distribution shift.

C. Model Architecture

We evaluate nine classifiers: six baselines (EfficientNet-B0 [44], MobileNetV3-Large [43], DenseNet-121 [42], ResNet-18, ResNet-50 [41], ConvNeXt-Tiny [45]) and three custom models (DraxNet, Drax-MobileNetV3-Large, FlipR). Full architecture of custom models are listed in Appendix A.

1) *DraxNet*: DraxNet (16.5M parameters) modifies a ResNet-18 backbone so the fourth stage consists of two residual blocks, each with a custom mixer (*DraxBlock*) inserted between its two convolutions. All other components are unchanged. *DraxBlock* consists of a ConvNeXt local convolutional branch followed by an optional self-attention branch, fused via stochastic-depth residuals. DraxNet, short for Dynamic Residual Attention eXchange, reserves the heavier mixer for the final, lowest-resolution stage, adding attention capacity with fewer parameters than a full attention backbone.

2) *Drax-MobileNetV3-Large*: Drax-MobileNetV3-Large (4.8M parameters) applies the DraxBlock to a parameter-efficient backbone. The MobileNetV3-Large feature extractor and head are unchanged, and a bottlenecked *DraxBlock* is inserted after the final feature stage. This tests whether the DraxBlock generalizes beyond a ResNet-style backbone.

3) *FlipR*: FlipR (2.8M parameters) inserts a *FlipR* block between the second and third stages of a ResNet-18 truncated to three stages, with the parameter-heavy fourth stage removed. The *FlipR* block computes the difference between each feature map and its horizontally flipped counterpart, smoothed by average pooling to account for imperfect anatomical symmetry, then broadcasts a gated mask across all channels. The block is inspired by SymAttention [40], which observes that bilateral asymmetry in CXRs is a key indicator of TB. FlipR tests whether a lightweight symmetry gate can match heavier attention approaches.

D. Model Training

All models are trained from scratch with a fixed random seed, batch size 16, 512×512 input, 100 epochs, Adam optimizer at a fixed learning rate of 10^{-3} , and unweighted cross-entropy loss. No hyperparameter tuning was done. The checkpoint with the lowest validation loss is retained and evaluated on the held-out test fold.

E. Results

Classification results are reported in Table II.

FlipR achieves comparable performance to the top models despite using the fewest parameters. The top four models (FlipR, EfficientNet-B0, MobileNetV3-Large, Drax-MobileNetV3-Large) differ by within $\pm 1\%$ on accuracy and F1. FlipR achieves high performance with the smallest parameter count (2.8M against 4.0 – 4.8M for the other three), consistent with the SymFormer [40] observation that bilateral asymmetry is a strong indicator of TB.

All models meet WHO targets for TB triage. Sensitivity ranges from 92.50 to 96.67%, above the WHO target of 90% sensitivity. Specificity ranges from 97.98 to 99.91%, also above the WHO target of 70% specificity [2]. Sensitivity is more clinically important than specificity in screening, because a missed TB case risks untreated transmission while a false positive triggers only unnecessary confirmatory testing. MobileNetV3-Large achieves the highest sensitivity, missing only 3.3% of TB cases.

The high-specificity, lower-sensitivity profile is likely attributable to TBX11K rather than model architecture. All models display high specificity and lower sensitivity, consistent with results from models in related literature trained on TBX11K. The TBX11K dataset is constructed to reward high specificity because the common failure mode on other TB datasets is over-predicting TB and achieving near-zero specificity [40].

Lightweight models are preferred for TB screening on both performance and deployment grounds. The top four models (FlipR, EfficientNet-B0, MobileNetV3-Large, Drax-MobileNetV3-Large) match or exceed every larger model in accuracy and F1 with lower parameters. Due to a modest TBX11K training set, a minority TB class, and no class-imbalance handling, larger networks appear to overfit the majority healthy and sick non-TB classes rather than learn discriminative TB features. TB screening also runs in resource-limited settings often without a dedicated GPU, so a lightweight model is preferable in deployment.

TABLE II. TUBERCULOSIS CLASSIFICATION RESULTS

Model	AUC _{TB}	Acc. (%)	Sens. (%)	Spec. (%)	Avg. Prec. (%)	Avg. Rec. (%)	Avg. F1 (%)	Params
FlipR	0.9984	98.97	94.17	99.91	99.01	97.70	98.34	2.8M
EfficientNet-B0	0.9995	98.89	95.00	99.65	98.29	97.87	98.07	4.0M
MobileNetV3-Large	0.9995	99.21	96.67	99.82	98.97	98.54	98.75	4.2M
Drax-MobileNetV3-Large	0.9989	98.81	95.00	99.65	98.23	97.81	98.02	4.8M
DenseNet-121	0.9975	98.57	92.50	99.56	97.81	96.97	97.38	7.0M
ResNet-18	0.9982	98.17	93.33	99.21	96.71	96.90	96.80	11.2M
DraxNet	0.9988	98.41	95.00	99.47	97.51	97.51	97.51	16.5M
ResNet-50	0.9978	97.70	92.50	99.47	96.95	96.33	96.63	23.5M
ConvNeXt-Tiny	0.9952	97.14	95.00	97.98	93.63	96.58	94.97	27.8M

AUC_{TB} is the area under the ROC curve for the TB class. Sensitivity is recall for the TB class. Specificity is recall for the non-TB class (healthy and sick non-TB combined). Precision, recall, and F1 are averaged across the three classes (healthy, sick non-TB, TB).

IV. TUBERCULOSIS LESION DETECTION

We frame TB lesion detection as an object detection task: the module localizes each TB region on the radiograph as a bounding box and classifies it, supplying the region-level evidence that the report generation module later renders as text.

A. Dataset

Object detection uses the TBX11K bounding-box annotations, with each region labeled active TB or latent TB. Dataset composition is shown in Table III.

Active TB denotes ongoing, transmissible infection requiring treatment. Latent TB denotes prior, inactive infection that is not contagious but may reactivate. As Figure 2 shows, active TB is more visually salient and latent TB is more subtle. A CXR with an image-level label of TB may contain bounding box labels of only active-TB lesions (630 images), only latent-TB lesions (139 images), or both (30 images). This composition is consistent with real-world conditions [39].

Differentiation between active and latent TB is relatively understudied in TB CAD. In routine reporting, a radiologist may not always state whether a lesion is suspected as active or latent TB explicitly, instead inferring it from radiographic pattern, with consolidation and cavitation suggesting active disease and fibrotic linear opacities suggesting healed disease, alongside clinical and laboratory correlation [4]. Current literature understudies whether existing commercial CAD systems can differentiate latent from active TB [21].

TABLE III. TBX11K DATASET COMPOSITION (BOUNDING BOX LABELS) FOR LESION DETECTION

Annotation	Train	Val	Total
Active-TB boxes	724	248	972
Latent-TB boxes	178	61	239
Total boxes	902	309	1,211

Counts are individual bounding boxes, not images. The 799 TB images (599 train, 200 val) contain 1,211 boxes, since one image may hold several regions.

B. Model Architecture

We evaluate YOLO26 and a custom variant in which the detector's default feature extractor is replaced by the DraxNet backbone used in classification. The detection head and training protocol are unchanged, isolating the backbone's effect on detection.

C. Model Training

Object detection uses the same training configuration as the classification models (Section III). As in classification, the best checkpoint (selected by mAP) is retained and used for evaluation.

D. Results

TABLE IV. TUBERCULOSIS LESION DETECTION RESULTS

Model	Class	mAP ₅₀	mAP ₅₀₋₉₅
YOLO26	ATB	0.4511	0.2110
YOLO26	LTB	0.1070	0.0410
YOLO26 + DraxNet	ATB	0.5865	0.2748
YOLO26 + DraxNet	LTB	0.1017	0.0310

ATB = active TB; LTB = latent TB.

Detection results are reported in Table IV.

Active TB is more likely to be detected than latent TB. Both detectors localize active TB roughly four times more accurately than latent TB at mAP₅₀ and roughly five to nine times more accurately at mAP₅₀₋₉₅. This gap reflects the TBX11K class imbalance (active-TB regions outnumber latent-TB regions roughly 4:1) and the subtle appearance of latent-TB lesions. The TBX11K reference reports the same pattern across all detection methods, attributing it to imbalance [40], and it is consistent with human readers distinguishing latent TB less reliably than active TB [5]. The bias toward active TB also aligns with clinical triage priorities, because active TB requires urgent treatment.

The DraxNet backbone specializes toward active TB. YOLO26 with the DraxNet backbone raises active-TB mAP₅₀ from 0.4511 to 0.5865 (+0.1354) and mAP₅₀₋₉₅ from

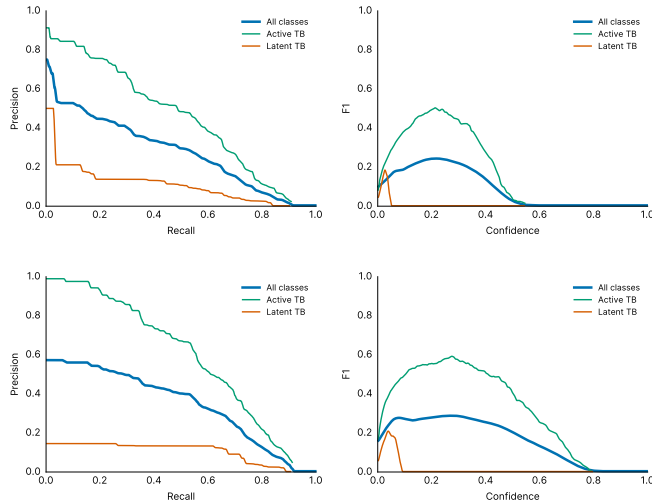


Fig. 3. Detection precision-recall (left) and F1-confidence (right) curves for YOLO26 (top) and YOLO26 with the DraxNet backbone (bottom). Both sustain high precision while recall saturates near 0.26, and the latent-TB curve collapses at low confidence, so localization functions as a high-precision rule-in cue rather than a standalone detector.

0.2110 to 0.2748 (+0.0638), while slightly reducing latent-TB (0.1070 $\rightarrow$ 0.1017 at mAP₅₀; 0.0410 $\rightarrow$ 0.0310 at mAP₅₀₋₉₅). The gain is driven almost entirely by precision (0.74 $\rightarrow$ 0.82). Recall is unchanged at roughly 0.26.

Recall, not precision, is the bottleneck. Both detectors localize confidently but find only about a quarter of the annotated TB regions. Figure 3 shows precision staying high while recall saturates near 0.26, and the latent-TB curve collapsing at low confidence for both detectors.

Localization is useful to confirm (rule-in) TB, but not to rule it out. The high-precision, low-recall profile indicates that a flagged lesion is likely correct and warrants focused radiologist review, but the model cannot reliably exclude TB because most annotated lesions are missed. Localization is therefore most reliable as supporting region-level evidence for the classifier and report generation modules, rather than as a standalone detector for TB. Clinicians using the model may reliably use it for rule-in diagnosis, not for rule-out diagnosis.

V. REPORT GENERATION

The report generation module turns the detector's output into a structured radiology-style report a reader can act on without interpreting a boxed region or heatmap. Example outputs are shown in Figures 2 and 4.

The module follows the two-stage detector-to-text decomposition of Danu et al. [11]. However, because the detector here emits a closed TB vocabulary rather than open findings, we replace their proprietary detector and fine-tuned language model with the public models of this paper and a deterministic module, reserving a generative model for future work that uses specific radiographic findings. For current use, template-based report generation suffices and does not require significant additional memory or compute for resource-constrained settings.

The module has two stages: an adapter that converts raw detector output into a validated structured record, and a

template-based report generator that renders that record as text. The second stage receives only a structured record of what the classification and object detection modules produced. This was designed to prioritize faithfulness to model findings, as image-conditioned vision-language models such as CXR-LLaVA [29] and LLaVA-Med [13], as well as related work using LLMs [11] may report findings the model detector never flagged.

A. Structured Adapter

The adapter maps classification and detection output to a fixed schema with a small, closed vocabulary. The image-level field carries the predicted label, the argmax over *healthy*, *sick non-TB*, and *TB*, together with its class probabilities. Each detected region carries three fields: its type (*active TB* or *latent TB*), a confidence band (*low*, *medium*, or *high*, thresholded at 0.5 and 0.8 on the detection score), and a location. The location is assigned by mapping the bounding-box centroid to one of nine cells in a 3×3 grid over the image (upper, middle, or lower row by left, center, or right column). An image with no detections yields an empty region list. The schema is validated before generation, so the report generator always receives well-formed input drawn from this vocabulary. The record states the type and coarse location of each region but no finer radiographic finding, because the detector labels regions only as active or latent TB.

All images are posteroanterior (PA) frontal scans. Hence, the standard radiological laterality convention is used, wherein the patient's right side appears on the image's left side and vice versa. For example, a lesion present in the image's upper left corner, is reported as an upper right lesion.

B. Template-Based Report Generation

Template-based report generation is a structured template over the closed schema. It renders the record as a short report in three labeled sections, mirroring the findings-to-impression structure of a radiology report [54]: a numbered FINDINGS list with one TB region per line including lesion type, model confidence band, and lesion zone, an IMPRESSION stating the screening result in clinical wording, and a RECOMMENDATION based on the predicted class, such as referral for bacteriological confirmation when TB is indicated.

The report describes only the detected TB regions and does not assess specific radiographic findings or cardiac, pleural, and skeletal observations. Due to lack of data, this is reserved for future work.

We chose a template over a generative model after evaluating both. An instruction-tuned language-model report generator (Phi-3.5-mini-instruct [53], run image-blind on the same record) was developed alongside this work, but for the current closed vocabulary it was found to offer identical output with no advantage. The output is a short, fixed-format report with one line per region, so a template matches it on faithfulness, readability, and the ordering of findings while removing any possibility of hallucination and model inference cost, an advantage in the resource-limited settings this pipeline targets. A generative model is needed only for future work when the detector reports a richer, open set of interacting findings.

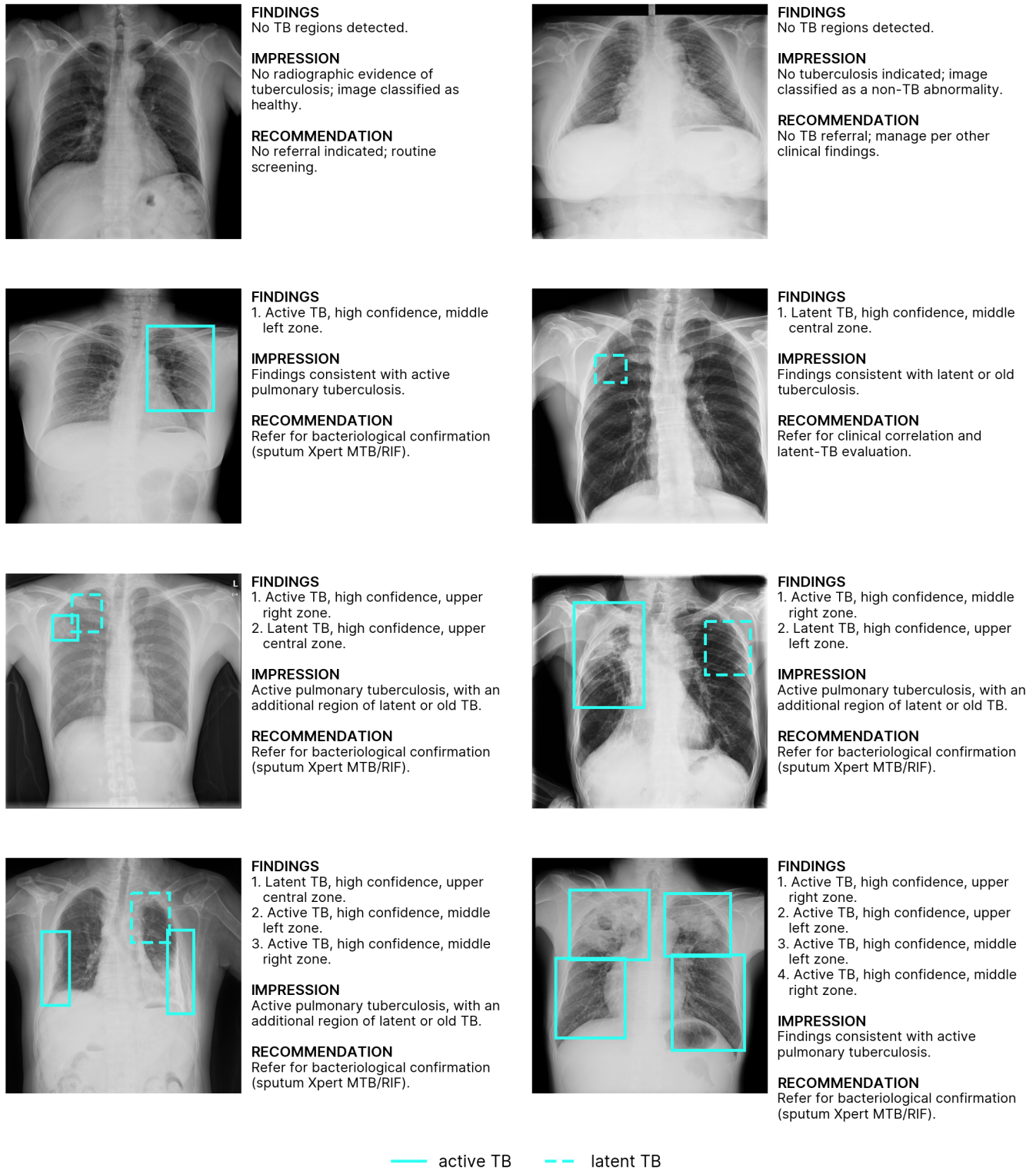


Fig. 4. Example generated reports on TBX11K scans.

C. Faithfulness by Construction

By design, report generation is faithful by construction, defined as the generated text asserting nothing not present in the structured record. This follows the faithfulness-versus-

factuality distinction drawn for abstractive summarization [48] and data-to-text evaluation [49], under which a generation is faithful if every entity it states is supported by the source, independent of whether the source is itself correct.

In previous experiments using Phi-3.5-mini-instruct for report generation, we evaluated the result for faithfulness with a deterministic verifier that parses a report for schema entities and flagging any the record does not contain: region type, location, or laterality absent from the record, a predicted label different from the record's, or an invented probability. The verifier follows an established approach in data-to-text generation, extracting the facts a generation asserts and checking them against the source records [50], and relates to the slot-error and semantic-accuracy metrics from dialogue generation that count output content not licensed by the input meaning representation [51], [52]. However, where those methods train an information-extraction or natural-language-inference model because the target text is open, the closed schema makes an exact deterministic match sufficient. Run over the full single-region surface of the schema, every combination of the two region types, three confidence bands, and nine zones together with the empty-region case for each class, the verifier reported 100% faithfulness. However, since the generative approach does not provide additional benefit over a template-based approach, the generative model is dropped to avoid hallucinations and reduce compute.

VI. DISCUSSION

A. Deployment Feasibility

Laptops are typically used to run CAD systems in community-based ACF, resulting in limited compute. CAD is also increasingly deployed offline where network connectivity is unreliable or absent [23], [21]. This mode already operates at scale, with more than 430,000 people screened by ultraportable X-ray and CAD between 2021 and 2023 [24]. Models for community-based ACF must be able to run locally on a CPU, without a dedicated GPU or a network connection.

Our lightweight models fall at the parameter scale shown in related literature to run on this class of hardware. Lightweight chest X-ray networks have been deployed on edge devices and CPUs with low latency, including ResNet-18, MobileNet, and EfficientNet-B0 on an embedded board [25] and a compact classifier on edge devices [26], and for TB specifically LightTBNets reports rapid, low-memory prediction designed for embedded use [46]. FlipR (2.8M parameters), Drax-MobileNetV3-Large (4.8M), EfficientNet-B0 (4.0M), and MobileNetV3-Large (4.2M) are at or below the size of those deployed networks, so they should run on the same devices. However, we did not conduct evaluation in real-world deployment to measure on-device latency or memory, and this remains future work.

B. Limitations

TB-specific samples are limited in public datasets. As of writing, TBX11K is the largest public chest X-ray dataset for tuberculosis with labeled bounding boxes, yet its TB subset remains small at 799 TB images and 1,211 TB lesion boxes. While general chest X-ray datasets exist, of these, TB-positive samples are few and insufficient to train from scratch. No public dataset contains TB-specific radiographs with reference radiographic reports, which is what a measurement of clinical correctness would require. This scarcity constrained the development and evaluation of report generation for clinical correctness.

Although report generation is faithful by construction, faithfulness to the detector is not clinical correctness. A report that faithfully restates an incorrect detection is still wrong, and only reference-based or reader evaluation can report this. However, due to the lack of a TB-specific reference dataset, this is left to future work.

The system's scope is limited to screening of non-pediatric patients since the radiographic presentation of pulmonary TB is age-dependent: adult reactivation tuberculosis appears in the upper lung zones, whereas pediatric primary tuberculosis appears in the lower lung zones. Hence, detection patterns learned from a predominantly adult distribution do not transfer to children. This is consistent with the WHO endorsement of CAD as a human-reader alternative only for patients aged 15 years and older [6].

C. Future Work

Future work seeks to evaluate the deployment efficiency of the pipeline in real-world use to benchmark inference latency, memory, and FLOPs on a target device.

Detection of full radiographic findings, such as cardiac size, pleural fluid, or non-TB abnormalities, as well as cardiac, pleural, and skeletal structures, are reserved for future work given additional data. Specific findings are more clinically meaningful for interpretation of the radiologic report, aligned with human reader interpretation of findings. This would also warrant use of a generative language model over a structured template for report generation, since synthesizing findings into a coherent report is no longer a fixed-format substitution but a genuine generation problem. This is demonstrated by recent chest X-ray systems that aggregate region-level findings into a report with an LLM [11], [55].

A reader study with radiologists and non-radiologist screeners is also recommended to evaluate the clinical correctness and coherence of the model-generated reports, particularly in comparison to human reader reports.

VII. CONCLUSION

TB CAD systems are highly accurate, however challenges remain in integration for use in community ACF with limited connectivity, compute, and with no radiologist present. This paper presented a lightweight TB screening pipeline that pairs lightweight detection models with deterministic, faithful-by-construction natural-language report generation of findings. The custom classifier, FlipR, achieves similar accuracy and F1 with the strongest baselines using the fewest parameters. The custom object detection model, DraxNet, improves active TB detection over YOLO26. Report generation provides findings in natural language for interpretation of ACF field teams and radiographers for patient referral for presumptive TB. TB-specific samples were found to be highly limited in public datasets, which constrained the design and evaluation of components. A full evaluation of deployment feasibility, extended radiographic findings, and a reader study with radiologists is left to future work.

ACKNOWLEDGMENTS

The authors would like to thank the University Research Council (URC) of Ateneo de Manila University for supporting

this work through a URC Standard Grant (URC 2025-20). The authors also thank Dr. Reynan B. Hernandez, radiologist at The Medical City and associate professor at the Ateneo School of Medicine and Public Health, for providing medical expertise that informed the clinical grounding of the study.

DECLARATION ON GENERATIVE AI

Generative AI tools were used to assist in drafting and preparing this manuscript. The conception of the research, methodology, and conclusions are the work of the authors.

APPENDIX A ARCHITECTURES OF CUSTOM MODELS

TABLE V. FLIPR ARCHITECTURE (RESNET-18 TRUNCATED TO THREE STAGES)

Stage	Output	Params
Stem (conv 7×7 , BN, ReLU, pool)	$64 \times H/4$	9,536
layer1 (2 basic blocks, 64)	$64 \times H/4$	147,968
layer2 (2 basic blocks, 128)	$128 \times H/8$	525,568
FlipR gate	$128 \times H/8$	129
layer3 (2 basic blocks, 256)	$256 \times H/16$	2,099,712
Head (avg-pool, FC $256 \rightarrow 3$)	3	771
Total		2,783,684

TABLE VI. DRAXNET ARCHITECTURE (RESNET-18, DRAXBLOCK IN STAGE 4)

Stage	Output	Params
Stem (conv 7×7 , BN, ReLU, pool)	$64 \times H/4$	9,536
layer1 (2 basic blocks, 64)	$64 \times H/4$	147,968
layer2 (2 basic blocks, 128)	$128 \times H/8$	525,568
layer3 (2 basic blocks, 256)	$256 \times H/16$	2,099,712
layer4 (2 DraxBlocks, 512)	$512 \times H/32$	13,699,072
Head (avg-pool, FC $512 \rightarrow 3$)	3	1,539
Total		16,483,395

TABLE VII. DRAX-MOBILENETV3-LARGE ARCHITECTURE

Stage	Output	Params
MobileNetV3-Large features	$960 \times H/32$	2,971,952
DraxBlock adapter (bottleneck 160)	$960 \times H/32$	575,200
Head (avg-pool, classifier $960 \rightarrow 1280 \rightarrow 3$)	3	1,233,923
Total		4,781,075

REFERENCES

- [1] World Health Organization, "Global Tuberculosis Report 2024," Geneva: WHO, 2024. [Online]. Available: <https://www.who.int/teams/global-tuberculosis-programme/tb-reports/global-tuberculosis-report-2024>
- [2] World Health Organization, "WHO consolidated guidelines on tuberculosis. Module 2: Screening – Systematic screening for tuberculosis disease," Geneva: WHO, 2021. [Online]. Available: <https://www.who.int/publications/i/item/9789240022676>
- [3] M. Pai, M. A. Behr, D. Dowdy, K. Dheda, M. Divangahi, C. C. Boehme, A. Ginsberg, S. Swaminathan, M. Spigelman, H. Getahun, D. Menzies, and M. Raviglione, "Tuberculosis," *Nature Reviews Disease Primers*, vol. 2, art. 16076, 2016, doi: 10.1038/nrdp.2016.76.
- [4] J. Burrill, C. J. Williams, G. Bain, G. Conder, A. L. Hine, and R. R. Misra, "Tuberculosis: A radiologic review," *RadioGraphics*, vol. 27, no. 5, pp. 1255–1273, 2007, doi: 10.1148/rg.275065176.
- [5] Y. R. Choi, S. H. Yoon, J. Kim, J. Y. Yoo, H. Kim, and K. N. Jin, "Chest radiography of tuberculosis: determination of activity using deep learning algorithm," *Tuberculosis and Respiratory Diseases*, vol. 86, no. 3, pp. 226–233, 2023, doi: 10.4046/trd.2023.0020.
- [6] World Health Organization, "WHO operational handbook on tuberculosis. Module 2: Screening," Geneva: WHO, 2021. [Online]. Available: <https://www.who.int/publications/i/item/9789240022614>
- [7] World Health Organization, "Use of computer-aided detection software for tuberculosis screening: WHO policy statement," Geneva: WHO, 2025. [Online]. Available: <https://www.who.int/publications/i/item/9789240110373>
- [8] H. K. Ahmad *et al.*, "Machine learning augmented interpretation of chest X-rays: A systematic review," *Diagnostics*, vol. 13, no. 4, art. 743, 2023. [Online]. Available: <https://pmc.ncbi.nlm.nih.gov/articles/PMC9955112/>
- [9] P. G. Anderson *et al.*, "Deep learning improves physician accuracy in the comprehensive detection of abnormalities on chest X-rays," *Scientific Reports*, vol. 14, 2024. [Online]. Available: <https://www.nature.com/articles/s41598-024-76608-2>
- [10] K. Kallianos, J. Mongan, S. Antani, T. Henry, A. Taylor, J. Abuya, and M. Kohli, "How far have we come? Artificial intelligence for chest radiograph interpretation," *Clinical Radiology*, vol. 74, no. 5, pp. 338–345, 2019. [Online]. Available: <https://doi.org/10.1016/j.crad.2018.12.015>
- [11] M. D. Danu, G. Marica, S. K. Karn, B. Georgescu, A. Mansoor, F. Ghesu, L. M. Itu, C. Suciu, S. Grbic, O. Farri, and D. Comaniciu, "Generation of radiology findings in chest X-ray by leveraging collaborative knowledge," *Procedia Computer Science (ITQM 2023)*, 2023. [Online]. Available: <https://arxiv.org/abs/2306.10448>
- [12] F. C. Ghesu, B. Georgescu, A. Mansoor, Y. Yoo, D. Neumann, P. Patel, R. S. Vishwanath, J. M. Balter, Y. Cao, S. Grbic, and D. Comaniciu, "Self-supervised learning from 100 million medical images," *arXiv preprint arXiv:2201.01283*, 2022. [Online]. Available: <https://arxiv.org/abs/2201.01283>
- [13] C. Li *et al.*, "LLaVA-Med: Training a large language-and-vision assistant for biomedicine in one day," *Advances in Neural Information Processing Systems*, 2023. [Online]. Available: <https://arxiv.org/abs/2306.00890>
- [14] N. Marquez *et al.*, "Performance of chest X-ray with computer-aided detection powered by deep learning-based artificial intelligence for tuberculosis presumptive identification during case finding in the Philippines," *BMC Global and Public Health*, 2025. [Online]. Available: <https://bmcbglobalpublichealth.biomedcentral.com/articles/10.1186/s44263-025-00198-y>
- [15] Qure.ai, "Qure.ai App User's Manual (qXR)." [Online]. Available: <https://documentation.qure.ai/>
- [16] Delft Imaging, "CAD4TB+." [Online]. Available: <https://delftimaging.com/solutions/ai-powered-medical-imaging-software/cad4tb/>
- [17] Lunit, "INSIGHT CXR." [Online]. Available: <https://www.lunit.io/en/products/cxr>
- [18] VinBrain, "DrAid for TB screening." [Online]. Available: <https://draid.ai/>
- [19] DeepTek, "Genki: AI for tuberculosis screening." [Online]. Available: <https://www.deeptek.ai/genki>
- [20] Infervision, "InferRead DR Chest: instruction for use," (hosted on the Stop TB Partnership Digital Health Technology Hub). [Online]. Available: <https://stoptb.org/assets/documents/dhthub/User%20Manual%20Software%20-%20InferRead%20DR%20Chest.pdf>
- [21] Stop TB Partnership, "Screening and triage for TB using computer-aided detection (CAD) technology and ultra-portable X-ray systems: A practical guide," Geneva, Switzerland, 2021. [Online]. Available: https://www.stoptb.org/sites/default/files/imported/page/Screening_and_Triage_for_TB_using_Computer-Aided_Detection_CAD_Technology_and_Ultra-portable_X-Ray_Systems-A_Practical_Guide_.pdf
- [22] FIND, "Digital chest radiography and computer-aided detection (CAD) solutions for tuberculosis diagnostics: Technology landscape

- analysis,” Geneva, Switzerland, 2021. [Online]. Available: https://www.finddx.org/wp-content/uploads/2023/02/20210401_technology_landscape_computer_aided_tb_FV_EN.pdf
- [23] Stop TB Partnership, “Ultra-portable digital X-ray systems,” Geneva, Switzerland. [Online]. Available: <https://www.stoptb.org/what-we-do/accelerate-tb-innovations/introducing-new-tools-project/ultra-portable-digital-x-ray-systems>
- [24] J. Min, J. Halton, C. Villegas, A. Chua, C. Hewison, and S. Deborggraeve, “Scanned: The global investments in computer-aided detection and ultraportable X-ray for tuberculosis,” *PLOS Global Public Health*, vol. 5, no. 3, e0004232, 2025, doi: 10.1371/journal.pgph.0004232.
- [25] E. Benmalek, W. Rhalem, A. Jbabi *et al.*, “Edge-based real-time diagnosis of pediatric pneumonia using lightweight CNNs and chest X-rays,” *Research on Biomedical Engineering*, vol. 41, art. 60, 2025, doi: 10.1007/s42600-025-00437-z.
- [26] S. Chauhan *et al.*, “Detection of COVID-19 using edge devices by a light-weight convolutional neural network from chest X-ray images,” *BMC Medical Imaging*, vol. 24, art. 1, 2024, doi: 10.1186/s12880-023-01155-7.
- [27] H. Chen, C. Gomez, C.-M. Huang, and M. Unberath, “Explainable medical imaging AI needs human-centered design: guidelines and evidence from a systematic review,” *npj Digital Medicine*, vol. 5, no. 156, 2022. [Online]. Available: <https://www.nature.com/articles/s41746-022-00699-2>
- [28] K. Borys, Y. A. Schmitt, M. Nauta, C. Seifert, N. Krämer, C. M. Friedrich, and F. Nensa, “Explainable AI in medical imaging: An overview for clinical practitioners – Beyond saliency-based XAI approaches,” *European Journal of Radiology*, vol. 162, art. 110786, 2023. [Online]. Available: <https://doi.org/10.1016/j.ejrad.2023.110786>
- [29] S. Lee, J. Youn, H. Kim, M. Kim, and S. H. Yoon, “CXr-LLaVA: a multimodal large language model for interpreting chest X-ray images,” *European Radiology*, vol. 35, 2025. [Online]. Available: <https://link.springer.com/article/10.1007/s00330-024-11339-6>
- [30] I. Hartsock and G. Rasool, “Vision-language models for medical report generation and visual question answering: a review,” *Frontiers in Artificial Intelligence*, vol. 7, art. 1430984, 2024. [Online]. Available: <https://pmc.ncbi.nlm.nih.gov/articles/PMC11611889/>
- [31] S. Vijayan *et al.*, “Implementing a chest X-ray artificial intelligence tool to enhance tuberculosis screening in India: Lessons learned,” *PLOS Digital Health*, vol. 2, no. 12, e0000404, 2023. [Online]. Available: <https://journals.plos.org/digitalhealth/article?id=10.1371/journal.pdig.0000404>
- [32] A. L. Innes *et al.*, “Computer-aided detection for chest radiography to improve the quality of tuberculosis diagnosis in Vietnam’s district health facilities: An implementation study,” *Tropical Medicine and Infectious Disease*, vol. 8, no. 11, art. 488, 2023. [Online]. Available: <https://doi.org/10.3390/tropicalmed8110488>
- [33] J. Creswell *et al.*, “Early user perspectives on using computer-aided detection software for interpreting chest X-ray images to enhance access and quality of care for persons with tuberculosis,” *BMC Global and Public Health*, vol. 1, no. 1, art. 30, 2023. [Online]. Available: <https://bmcbglobalpublichealth.biomedcentral.com/articles/10.1186/s44263-023-00033-2>
- [34] A. J. Scott *et al.*, “Diagnostic accuracy of computer-aided detection during active case finding for pulmonary tuberculosis in Africa: A systematic review and meta-analysis,” *Open Forum Infectious Diseases*, vol. 11, no. 2, ofae020, 2024. [Online]. Available: <https://doi.org/10.1093/ofid/ofae020>
- [35] Z.-L. Han *et al.*, “A systematic review and meta-analysis of artificial intelligence software for tuberculosis diagnosis using chest X-ray imaging,” *Journal of Thoracic Disease*, vol. 17, no. 5, pp. 3223–3237, 2025, doi: 10.21037/jtd-2025-604.
- [36] I. F. Ihongbe, S. Fouad, T. F. Mahmoud, A. Rajasekaran, and A. Bhatia, “Evaluating explainable artificial intelligence (XAI) techniques in chest radiology imaging through a human-centered lens,” *PLOS ONE*, vol. 19, no. 10, e0308758, 2024. [Online]. Available: <https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0308758>
- [37] S. Gaube *et al.*, “Care to explain? AI explanation types differentially impact chest radiograph diagnostic performance and physician trust in AI,” *Radiology*, vol. 313, no. 1, e233261, 2024. [Online]. Available: <https://pubs.rsna.org/doi/10.1148/radiol.233261>
- [38] D. Hua *et al.*, “Towards human-AI collaboration in radiology: A multidimensional evaluation of the acceptability of AI for chest radiograph analysis in supporting pulmonary tuberculosis diagnosis,” *JAMIA Open*, vol. 8, no. 1, ooae151, 2025. [Online]. Available: <https://academic.oup.com/jamiaopen/article/8/1/ooae151/8002370>
- [39] Y. Liu, Y.-H. Wu, Y. Ban, H. Wang, and M.-M. Cheng, “Rethinking computer-aided tuberculosis diagnosis,” in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2020, pp. 2646–2655. [Online]. Available: <https://mmcheng.net/tb/>
- [40] Y. Liu, Y.-H. Wu, S.-C. Zhang, L. Liu, M. Wu, and M.-M. Cheng, “Revisiting computer-aided tuberculosis diagnosis,” *IEEE Trans. Pattern Analysis and Machine Intelligence*, 2023. [Online]. Available: <https://arxiv.org/abs/2307.02848>
- [41] K. He, X. Zhang, S. Ren, and J. Sun, “Deep residual learning for image recognition,” in *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 770–778. [Online]. Available: <https://arxiv.org/abs/1512.03385>
- [42] G. Huang, Z. Liu, L. van der Maaten, and K. Q. Weinberger, “Densely connected convolutional networks,” in *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, 2017, pp. 4700–4708. [Online]. Available: <https://arxiv.org/abs/1608.06993>
- [43] A. Howard *et al.*, “Searching for MobileNetV3,” in *Proc. IEEE/CVF Int. Conf. Computer Vision (ICCV)*, 2019, pp. 1314–1324. [Online]. Available: <https://arxiv.org/abs/1905.02244>
- [44] M. Tan and Q. V. Le, “EfficientNet: Rethinking model scaling for convolutional neural networks,” in *Proc. Int. Conf. Machine Learning (ICML)*, 2019, pp. 6105–6114. [Online]. Available: <https://arxiv.org/abs/1905.11946>
- [45] Z. Liu, H. Mao, C.-Y. Wu, C. Feichtenhofer, T. Darrell, and S. Xie, “A ConvNet for the 2020s,” in *Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR)*, 2022, pp. 11976–11986. [Online]. Available: <https://arxiv.org/abs/2201.03545>
- [46] D. Capellán-Martín, J. J. Gómez-Valverde, D. Bermejo-Peláez, and M. J. Ledesma-Carbayo, “A lightweight, rapid and efficient deep convolutional network for chest X-ray tuberculosis detection,” in *Proc. IEEE 20th Int. Symp. Biomedical Imaging (ISBI)*, 2023. [Online]. Available: <https://arxiv.org/abs/2309.02140>
- [47] M. Owda, A. Abumihisan, A. Y. Owda, and M. Abumohsen, “A lightweight hybrid deep learning model for tuberculosis detection from chest X-rays,” *Diagnostics*, vol. 15, no. 24, art. 3216, 2025, doi: 10.3390/diagnostics15243216.
- [48] J. Maynez, S. Narayan, B. Bohnet, and R. McDonald, “On faithfulness and factuality in abstractive summarization,” in *Proc. 58th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2020, pp. 1906–1919. [Online]. Available: <https://arxiv.org/abs/2005.00661>
- [49] B. Dhingra, M. Faruqui, A. Parikh, M.-W. Chang, D. Das, and W. Cohen, “Handling divergent reference texts when evaluating table-to-text generation,” in *Proc. 57th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2019, pp. 4884–4895. [Online]. Available: <https://arxiv.org/abs/1906.01081>
- [50] S. Wiseman, S. M. Shieber, and A. M. Rush, “Challenges in data-to-document generation,” in *Proc. Conf. Empirical Methods in Natural Language Processing (EMNLP)*, 2017, pp. 2253–2263. [Online]. Available: <https://arxiv.org/abs/1707.08052>
- [51] T.-H. Wen, M. Gašić, N. Mrkšić, P.-H. Su, D. Vandyke, and S. Young, “Semantically conditioned LSTM-based natural language generation for spoken dialogue systems,” in *Proc. Conf. Empirical Methods in Natural Language Processing (EMNLP)*, 2015, pp. 1711–1721. [Online]. Available: <https://arxiv.org/abs/1508.01745>
- [52] O. Dušek and Z. Kasner, “Evaluating semantic accuracy of data-to-text generation with natural language inference,” in *Proc. 13th Int. Conf. Natural Language Generation (INLG)*, 2020, pp. 131–137. [Online]. Available: <https://arxiv.org/abs/2011.10819>
- [53] M. Abdin *et al.*, “Phi-3 technical report: A highly capable language model locally on your phone,” *arXiv preprint arXiv:2404.14219*, 2024. [Online]. Available: <https://arxiv.org/abs/2404.14219>
- [54] L. Zhang, M. Liu, L. Wang, Y. Zhang, X. Xu, Z. Pan, Y. Feng, J. Zhao, L. Zhang, G. Yao, X. Chen, and X. Xie, “Constructing a large language model to generate impressions from findings in radiology reports,” *Radiology*, vol. 312, no. 3, e240885, 2024. [Online]. Available: <https://doi.org/10.1148/radiol.240885>

- [55] W.-J. Noh, S.-W. Pi, and B.-D. Lee, "Hybrid framework for lesion-aware, clinically coherent chest X-ray report generation using contrastive learning and large language models," *Scientific Reports*, 2026. [Online]. Available: <https://pmc.ncbi.nlm.nih.gov/articles/PMC12868645/>